(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

SIXTH SEMESTER [B.TECH] JULY 2023

Paper Code: ETCS-302 ***Subject: Compiler Design***

Time: 3 Hours ***Maximum Marks: 75***

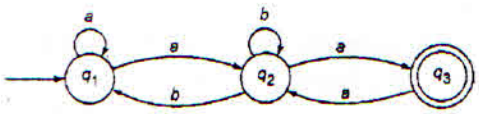
Note: Attempt five questions including Q. No. 1 which is compulsory. Assume missing data. Select one question from each unit.

Q1. Answer the following questions: **(5*5=25)**

a) What is Backpatching? Explain with the help of example.
b) Distinguish between Compiler and Interpreter briefly.
c) Explain various principle sources of Code Optimization.
d) What is DAG? What are the advantages of DAG?
e) What is the role and various functions of Lexical analyser? Explain briefly.

UNIT-I

Q2. a) Discuss the architecture of Compiler with its phases in details **(7.5)**
b) Consider the transition diagram **(5)**



Convert the above finite automata into the regular expression.

Q3. a) What is bottom-up parsing? How to construct the SLR parsing table? Explain with the help of an example. **(6.5)**

b) For the Grammar give below: **(6)**
E-> TE'
E'->+TE'/ε
T->FT'
F->*FT'/ε
F->(E)/id
Construct the Predictive Parsing Table. Whether this grammar is LL(1) or not?

UNIT-II

Q4. a) What is syntax directed translation? How the syntax directedtranslation schemes aredescribed. Explain with the help of example. **(6)**

b) What is intermediate code? What are the various ways to represent the intermediate code? Explain postfix notation, parse trees and syntax trees, three address code, quadruples, and triples with the help of examples. **(6.5)**

Q5. a) Describe the language for specifying Lexical Analyzer with the help of example. **(6)**

b) Write a short note on: Type Checker, Type Conversion and Boolean Expressions. **(6.5)**

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First-Term Examination

Sixth Semester (B.Tech) April-2022

Paper Code: ETCS-302 Subject: Compiler Design

Time: 1 Hr 30 Mins Max. Marks: 30

Note: Q1 is compulsory and attempt any two questions out of remaining three.

Q1 Attempt all five parts (5*2 =10)

a) What is a cross compiler and how it is different from native compiler?

b) What is the need for separating the parser from scanner?

c) Find leftmost derivation, rightmost derivation and derivation tree of a string 'aacbbcc' from following grammar productions {S→aAcB|Bds , B→aAcA|cAB|b , A→aB|aBc|a}.

d) What are the responsibilities of Loader, Linker and Assembler in the compiler environment?

e) Write steps involved in compilation of following sentence

pos = interest + rate * 60

Q2 Attempt both parts (2*5=10)

a) Remove indirect left recursion from the following grammar:

S → Aa | b

A→Ac | Sd | e

b) Prove that the following grammar is ambiguous and convert it to unambiguous one.

S-> S+S|S*S|(S)|a

Q 3 (1*10=10)

Write Algorithm to construct predictive parser. Design a predictive parser for the given grammar and show the processing of one string.

S→aABb

A→c|∈

B→d|∈

Q 4 Attempt both parts (2*5=10)

a) Draw the finite automata which can be used by lexical analyzer to recognize identifier, number and operators.

b) What is operator grammar? Write operator precedence parsing table for following grammar:

S→(L)|a

L→L ,S|S
